AI-Based Medical Representative

Visit Prioritization & Route Optimization System

ABSTRACT

Medical Representatives (MRs) are required to plan and manage multiple doctor visits within limited working hours, where doctor priority, availability, visit duration, and geographical distance can make manual planning inefficient. This project proposes an AI-based web application that automates Medical Representative Visit prioritization and daily route planning using Machine Learning and optimization techniques. The system processes historical doctor, appointment, and visit-related data to predict a priority score for planned doctor visits using a Random Forest model. The predicted priority, along with doctor availability, visit duration, MR working hours, and location constraints, is then provided to Google OR-Tools for Vehicle Routing Problem with Time Windows (VRPTW) based route optimization. The system generates an optimized sequence of doctor visits and provides role-based dashboards for Medical Representatives and managers. The proposed solution aims to reduce unnecessary travel, improve visit scheduling, prioritize important visits, and provide better visibility into field activities, enabling data-driven and efficient Medical Representative planning.

Keywords:

Medical Representative, Machine Learning, Visit Prioritization, Route Optimization, Random Forest, Google OR-Tools, Field Force Management

Group No: G19

Student Name:

Bhakti Gundecha – 3071

Parth Lokhande – 1020

Udaykiran Bitla – 3056

Internal Guide

Prof. Mrs. M. U. Buchade